

Biblatex Compatibility (hibib.dtx)

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Because metadata for legal citations is more structured and complex than for other citation systems, *Hereinafter* does not use Biblatex files as its primary input format. However, the `hibib` package provides a compatibility layer that uses Biblatex to process `.bib` files into *Hereinafter* data structures. The package also enables rudimentary production of `.bib` files from *Hereinafter* reference definitions.

```
\ProvidesPackage{hibib}[2021/02/26 Hereinafter to BibLaTeX conversion]
\RequirePackage{histrings}
\RequirePackage{etoolbox}
\RequirePackage[datamodel=hibl]{biblatex}
```

1 Initial Setup

There are two initial setup tasks that must be completed. First, the reference type names are not consistent between *Hereinafter* and Biblatex, so a mapping first needs to be established.

```
\def\hi@bib@map#1#2{%
  \namedef{hi@bib@kind@#1}{#2}%
  \namedef{hi@bib@hikind@#2}{#1}%
  \csletcs{def#2}{def#1}%
}
\hi@bib@map{jrnart}{article}
\hi@bib@map{website}{misc}
\hi@bib@map{website}{online}
\hi@bib@map{case}{jurisdiction}
\hi@bib@map{procart}{inproceedings}
```

Converts a *Hereinafter* reference type to a biblatex reference type.

```
\def\hi@bib@bibkind#1{%
  \@ifundefined{hi@bib@kind@#1}{#1}{\csname hi@bib@kind@#1\endcsname}%
}
```

Converts a biblatex reference type to a *Hereinafter* reference type.

```
\def\hi@bib@hiakind#1{%
  \@ifundefined{hi@bib@hikind@#1}{#1}{\csname hi@bib@hikind@#1\endcsname}%
}
```

Second, Biblatex only retains references that are actually cited in a document. Accordingly, we need to signal that every cited reference should be retained, by issuing a `\nocite` command.

Internally, we patch `\hi@pse@sigvolref`, which is called every time a reference is used in a citation. The hook below informs biblatex of the references used by adding a `\nocite` command, indicating to Biblatex that the reference should be included in the `.bcf` file.

```
\apptocmd\hi@pse@sigvolref{%
  \expandafter\ifx\csname fc@\this@case\endcsname\fc@this \else
    \expandafter\nocite\expandafter{\@this@case}%
  \fi
}{}{}%
```

2 Producing Biblatex Files

\hiBibFile The `\hiBibFile{<file>}` macro initiates production of a Biblatex file. The command should be called before any *Hereinafter* references are defined.

Internally, we patch `\hi@param@read` (described in `refs.dtx`) to collect the reference's parameters and then output the reference definition to the bibliography file.

```
\def\hiBibFile#1{%
  \newwrite\tf@bib
  \immediate\openout \tf@bib #1.bib\relax
  \pretocmd\hi@param@read{\let\hi@bib@paramlist\empty}{}{}
  \apptocmd\hi@param@read{\hi@bib@output}{}{}
}
\let\tf@bib\relax
```

If `\hiBibFile` has been executed, then the `\hi@bib@output` macro below is executed after the parameters for a reference have been read. The macro generates a Biblatex-style reference definition and writes it to the bibliography file defined by `\hiBibFile`.

```
\def\hi@bib@output{%
  \ifx\tf@bib\relax\else
    \let\reserved@b\empty
    \expand{\unbrace{\etfor\reserved@a:=}}\hi@bib@paramlist i\do{%
      \ifundefined\hi@bib@p@reserved@a{}{}%
      \protected@edef\reserved@b{%
        \reserved@b
        \space\space\reserved@a={%
          \csname hi@bib@p@reserved@a \endcsname
        },^^J%
      }%
      \expandafter\let\csname hi@bib@p@reserved@a\endcsname \relax
    }%
  }%
  \immediate\write\tf@bib{%
    @\expandafter\hi@bib@bibkind\expandafter{\hi@kv@kind}{%
      \ethis@case,^^J%
      \reserved@b
    }^^J%
  }%
}
\fi
}
```

3 Reference Parameters

`\hi@bib@paramlist` is a list of parameters to be included when printing a bibliography reference. This is set to `\empty` inside `\hi@param@read` (see `\hiBibFile` above) and then augmented.

```
\let\hi@bib@paramlist\relax
```

We patch each `\KV@hi@<param>` macro, to execute its normal operations and then call `\hi@bib@doparam`. The `\hi@params` macro lists every parameter with a `\do` command.

```
\def\do#1#2{%
  \def\reserved@a#1{%
    \@namedef{KV@hi@#1}###1{%
      % In here, |#1| is the param name, |##1| is the prior macro
      % definition of the param, and |###1| is the variable name for the
      % argument the param's new macro will take.
      ##1\hi@bib@doparam{#1}{###1}%
    }%
  }%
  \expand\reserved@a{\csname KV@hi@#1\endcsname{##1}}{ii}%
}
\hi@params
```

Upon setting a parameter in a reference definition, this macro performs the work necessary to have the parameter saved to the Bibtex output. If a macro `\hi@bib@param@<param>` is defined, then that macro is executed with the parameter value as the argument. Otherwise, the default `\hi@bib@saveparam` macro is run. #1 is the parameter name; #2 the given value.

```
\def\hi@bib@doparam#1#2{%
  %
  % Only run if we are saving parameters
```

```

%
\ifx\hi@bib@paramlist\relax \else
\ifundefined{hi@bib@param@#1}{%
\ifblank{#2}{%
%
% For blank parameters, if there is a default value then use
% that. Otherwise do not output to the .bib file (because biber
% will ignore it anyway).
%
\ifundefined{KV@hi@#1@default}{}{%
\expand{\hi@bib@saveparam{#1}}{%
\csname kv@hi@#1@default\endcsname
}{i}%
}%
}%
\hi@bib@saveparam{#1}{#2}%
}%
}%
\csname hi@bib@param@#1\endcsname{#2}%
}%
\fi
}

```

Save a given value for the parameter by appending it to `\hi@bib@paramlist`, and defining a new macro `\hi@bib@p@#1` with the parameter's value. #1 is the parameter; #2 the value.

```

\def\hi@bib@saveparam#1#2{%
\def\reserved@a{#2}%
\expandafter\edef\csname hi@bib@p@#1\endcsname{%
\expandafter\strip@prefix\meaning\reserved@a
}%
\addto\macro\hi@bib@paramlist{{#1}}%
}

```

These are utility macros for changing how parameters are output to the Bibtex file.
Ignore a particular parameter.

```

\def\hi@bib@ignore#1{\namedef{hi@bib@param@#1}#1{}}%

```

Rename a parameter.

```

\def\hi@bib@rename#1#2{%
\namedef{hi@bib@param@#1}#1{%
\hi@bib@saveparam{#2}{#1}%
}%
}

```

By default, the Bibtex parameter's value is the user's given value, regardless of *Hereinafter's* processing. This macro directs the output to be the post-processed value.

```

\def\hi@bib@postval#1{%
\namedef{hi@bib@param@#1}#1{%
\expand{\hi@bib@saveparam{#1}}{\csname hi@kv@#1\endcsname}{i}%
}%
}

```

Like `\hi@bib@postval` but allows changing the output parameter name (#2).

```

\def\hi@bib@postval@rename#1#2{%
\namedef{hi@bib@param@#1}#1{%
\expand{\hi@bib@saveparam{#2}}{\csname hi@kv@#1\endcsname}{i}%
}%
}

```

For a parameter list #1, adds #2 to the list.

```

\def\hi@bib@list@add#1#2{%
\ifundefined{hi@bib@p@#1}{%
\hi@bib@saveparam{#1}{#2}%
}%
\expandafter\addto\macro\csname hi@bib@p@#1\endcsname{ and #2}%
}%
}

```

For parameter list #1, adds name #2 to the list.

```

\def\hi@bib@name@add#1#2{%
\hi@namesplit{#2}{\hi@bib@name@addsplit{#1}}%
}

```

3.1 Name Parameters

We have to mangle the name parts in order to be approximately compatible with Biblatex expectations. The translation is as follows:

- Given name $\Rightarrow$ Biblatex given name
- Family name + suffix $\Rightarrow$ Biblatex family name
- Institution $\Rightarrow$ Biblatex suffix

```
\def\hi@bib@name@addsplit#1#2#3#4#5#6{%
  \def\reserved@a{given={#2}}%
  \ifblank{#3}{%
    % I don't have any way of doing a name suffix with no family name, so we
    % just pretend that the suffix is the family name
    \notblank{#4}{\appto\reserved@a{, family={#4}}}{}}%
  }{%
    \ifblank{#4}{\appto\reserved@a{, family={#3}}}{%
      \appto\reserved@a{, family={#3 {#4}}}%
    }%
  }%
  \notblank{#5}{\appto\reserved@a{, "suffix={#5}"}{}}%
  \@expand{\hi@bib@list@add{#1}}\reserved@a i%
  \ifblank{#6}{\hi@bib@list@add{#1}{others}}%
}
```

3.2 Date Parameters

Deal with a date parameter. This sets up just the .bib output. #1 is the prefix to the date parameter name.

```
\def\hi@bib@date#1{%
  \hi@bib@ignore{#1date}% The date parameter is an alias
  \hi@bib@rename{#1year}{#1datetext}%
}
\def\hi@bib@month#1{%
  \ifcase#1\or
    Jan.\or Feb.\or Mar.\or Apr.\or May\or June\or
    July\or Aug.\or Sept.\or Oct.\or Nov.\or Dec.\or
    13\or14\or15\or16\or17\or18\or19\or20\or % Next is 21
    Spring\or Summer\or Fall\or Winter\or
    Spring\or Summer\or Fall\or Winter\or % Northern hemisphere
    Spring\or Summer\or Fall\or Winter\or % Southern hemisphere
    Spring\or Summer\or Fall\or Winter\or % Southern hemisphere
    Q1\or Q2\or Q3\or Q4\else ???%
  \fi
}
```

3.3 Specific Parameter Mappings

```
\def\hi@bib@param@agency#1{%
  \hi@bib@saveparam{agency}{#1}%
  \expandafter\let\csname hi@bib@p@court\endcsname\relax
}
\hi@bib@ignore{broadcaster}
\hi@bib@ignore{bill}
\hi@bib@ignore{cite}
\hi@bib@ignore{comment}
\hi@bib@ignore{docname}
\hi@bib@ignore{jcite}
\hi@bib@ignore{kind}
\hi@bib@ignore{issuer}
\hi@bib@ignore{journal}
\hi@bib@ignore{parse}
\hi@bib@ignore{parties}
\hi@bib@ignore{producer}
\hi@bib@ignore{publiclaw}
\hi@bib@ignore{publno}
\hi@bib@ignore{sameparties}
\hi@bib@ignore{serial}
\hi@bib@ignore{series}
\hi@bib@ignore{slipop}
\hi@bib@ignore{sponsor}
\hi@bib@ignore{src}
\hi@bib@ignore{state}
\hi@bib@ignore{volume}
\hi@bib@ignore{revparties}
```

```

\hi@bib@rename{hyphenation}{hyphenate}
\hi@bib@rename{name}{title}
\hi@bib@rename{page}{pages}
\hi@bib@rename{rep}{journaltitle}
\hi@bib@rename{inline}{shorttitle}
\hi@bib@date{}
\hi@bib@date{orig}
\hi@bib@date{issue}
\def\hi@bib@param@author#1{\hi@bib@name@add{author}{#1}}
\def\hi@bib@param@instauth#1{\hi@bib@list@add{author}{#1}}
\def\hi@bib@param@editor#1{\hi@bib@name@add{editor}{#1}}
\def\hi@bib@param@insted#1{\hi@bib@list@add{editor}{#1}}
\def\hi@bib@param@to#1{\hi@bib@name@add{to}{#1}}
\def\hi@bib@param@instto#1{\hi@bib@list@add{to}{#1}}
\def\hi@bib@param@paren#1{\hi@bib@list@add{paren}{#1}}
\def\hi@bib@param@country#1{\hi@bib@list@add{country}{#1}}
\hi@bib@postval{citation}
\hi@bib@postval{in}
\hi@bib@postval{reprinted}
\def\hi@bib@param@vol#1{%
  \find@in{.}{#1}{%
    \@tworun{\hi@bib@saveparam{volume}}{\hi@bib@saveparam{issue}}%
  }{\hi@bib@saveparam{volume}{#1}}%
}

```

4 Reading a Biblalex File

To use a Biblalex file for input, include it using the usual `\addbibresource` command. The `hibib` package patches Biblalex's input commands so that each time a reference is read, a corresponding *Hereinafter* reference is defined.

Note that for Biblalex to operate, the `.tex` file must be compiled first to produce a `.bcf` file, which is then processed through the `biber` program to produce a `.bbl` file that actually provides reference definitions. This package requires the same process. The document with citation commands must be compiled first, producing warnings about unknown references for every citation. Then `biber` may be called and the document recompiled.

Biblalex does not offer a way to collect all the parameters it reads from the `.bbl` file. So first we patch the necessary methods. These will construct a list `\hi@bib@bblparams` to contain every parameter in the bibliography entry.

```

\apptocmd\blx@bbl@entry{\let\hi@bib@bblparams\empty}{}{}
\apptocmd\blx@bbl@fielddef{\listadd\hi@bib@bblparams{#1}}{}{}
\apptocmd\blx@bbl@namedef{\listadd\hi@bib@bblparams{#1}}{}{}
\apptocmd\blx@bbl@listdef{\listadd\hi@bib@bblparams{#1}}{}{}

```

Now install a hook performed at the time the `.bbl` file is read. `\AtDataInput` executes after a `.bbl` file entry is read, so the parameters are stored in Biblalex's internal memory.

At that point, `\hi@param@read` is redefined to iterate over all the parameters found in `\hi@bib@bblparams`, and process each one using `\hi@bib@setparams@do`. Next, the reference definer macro is constructed and executed. This will call `\hi@param@read`, thereby defining a *Hereinafter* reference using Biblalex's parameters.

```

\AtDataInput{%
  \begingroup
  \def\hi@param@read#1{%
    \forlistloop\hi@bib@setparams@do\hi@bib@bblparams
  }%
  \hi@bib@writeconv{%
    \expandafter\string\csname
      def\usefield\hi@bib@hiakind{entrytype}%
    \endcsname{%
      \thefield{entrykey}%
    }{\charlb
  }%
  \edef\reserved@a{%
    \expandafter\noexpand\csname def\csfield{entrytype}\endcsname
    {\thefield{entrykey}}}%
  \reserved@a
  \hi@bib@writeconv{%\@charrb}%
}

```

```

\endgroup
}

```

`\hi@bib@setparams@do` is called for each Biblatex parameter that was recorded, during the execution of `\hi@param@read` as a reference is being defined. It does one of the following depending on the parameter it finds:

- If a special processing macro `\hi@bib@bbl@⟨param⟩` is defined, then that macro is executed with no arguments.
- If no default macro exists of the form `\KV@hi@⟨param⟩`, then nothing happens.
- If the field is undefined in the .bbl file, a warning is given.
- Otherwise, calls `\usefield` to get the Biblatex field value, and executes `\hi@bib@sethiparam` to transfer the value to *Hereinafter*.

```

\def\hi@bib@setparams@do#1{%
  \ifundefined{hi@bib@bbl@#1}{%
    \ifundefined{KV@hi@#1}{%
      \iffieldundef{#1}{%
        \PackageWarning\hi@pkgname{%
          Field #1 in .bbl file not found%
        }%
      }{%
        \usefield\hi@bib@sethiparam{#1}{#1}%
      }%
    }%
  }{%
    \csname hi@bib@bbl@#1\endcsname
  }%
}

```

Sets a *Hereinafter* parameter to a given value. Note that this is backwards in terms of parameters: #1 is the *value*, and #2 is the parameter name. This macro also write the parameter to a file if requested.

```

\def\hi@bib@sethiparam#1#2{%
  \csname KV@hi@#2\endcsname{#1}%
  \hi@bib@writeconv[ ]{\space\space#2={#1},}%
}

```

Macros of the form `\hi@bib@bbl@⟨param⟩` are special handlers for parameters from the .bbl file.

```

\def\hi@bib@bbl@paren{%
  \gdef\hi@bib@tmp{%
    \indexlist[bbparenprint]{paren}%
    \hi@bib@tmp
  }
  \DeclareIndexListFormat{bbparenprint}{%
    \gappto\hi@bib@tmp{%
      \hi@bib@sethiparam{#1}{paren}%
    }%
  }
}

```

4.1 Name Lists

These set up conversion of Biblatex's author lists to *Hereinafter* format.

```

\def\hi@bib@bbl@author{\hi@bib@bbl@namelist{author}{author}{instauth}}
\def\hi@bib@bbl@editor{\hi@bib@bbl@namelist{editor}{editor}{insted}}
\def\hi@bib@bbl@to{\hi@bib@bbl@namelist{to}{to}{instto}}
\def\hi@bib@bbl@namelist#1#2#3{%
  \def\hi@bib@bbl@hummac{#2}%
  \def\hi@bib@bbl@instmac{#3}%
  \ifandothers{#1}{%
    \gdef\hi@bib@maybeetal{et al.}%
  }{%
    \gdef\hi@bib@maybeetal{}%
  }%
  \gdef\hi@bib@tmp{%
    \indexnames[bbauthprint]{#1}%
    \hi@bib@tmp
  }
  \DeclareIndexNameFormat{bbauthprint}{%
    \ifdefvoid\namepartgiven{%
      % Institutional author.
      \edef\reserved@a{%
        \noexpand\hi@bib@sethiparam{%

```

```

\expandonce\namepartfamily
}{\hi@bib@bblinstmac}%
}%
}%
% Human author.
\edef\reserved@a{%
\noexpand\hi@bib@sethiparam{%
parts:
{\expandonce\namepartgiven}%
{\expandonce\namepartfamily}%
{\expandonce\namepartsuffix}%
{\expandonce\hi@bib@maybeetal}%
}{\hi@bib@bblhummac}%
}%
\global\let\maybeetal\@empty
}%
\expandafter\gappto\expandafter\hi@bib@tmp\expandafter{%
\reserved@a
}%
}

```

Convert Biblatex's delimiters to simple spaces.

```

\def\bibnamedelima{ }
\def\bibnamedelimb{ }
\def\bibnamedelimc{ }
\def\bibnamedelimd{ }
\def\bibnamedelimi{ }

```

4.2 Dates

This sets up date parameters for conversion from Biblatex to *Hereinafter*.

```

\def\hi@bib@bbldate#1{%
\@namedef{\hi@bib@bbl@#1datetext}{%
\usefield\hi@bib@sethiparam{#1datetext}{#1year}%
}%
\@namedef{\hi@bib@bbl@#1year}{%
\iffielddundef{#1datetext}{%
\edef\reserved@a{%
\noexpand\hi@bib@sethiparam{%
\hi@bib@bbldate@construct{#1}%
}{#1year}%
}\reserved@a
}%
}%
}%
\def\hi@bib@bbldate@construct#1{%
\csname if#1datecirca\endcsname{c. }{}%
\iffielddundef{#1month}{}%
\usefield\hi@bib@month{#1month}%
\iffielddundef{#1endmonth}{}%
\iffielddsequal{#1month}{#1endmonth}{}%
-\usefield\hi@bib@month{#1endmonth}%
}%
}%
\iffielddundef{#1day}{}%
\space
\csfield{#1day}%
\iffielddundef{#1endday}{}{-\csfield{#1endday}}%
,%
}%
\space
}%
\csfield{#1year}%
\iffielddundef{#1endyear}{}%
\iffielddsequal{#1year}{#1endyear}{}%
-\csfield{#1endyear}%
}%
}%
}%
\hi@bib@bbldate{
\hi@bib@bbldate{orig}
\hi@bib@bbldate{issue}

```

5 Saving Biblatex Entries in *Hereinafter* Format

\hiBibConvert It may be desirable to convert an existing Biblatex file into reference definition

commands compatible with *Hereinafter*, so that they can be further edited and refined for example. The process for doing this is unfortunately somewhat convoluted, and uses the command `\hiBibConvert` from this package.

To use it, construct a document like this:

```
\usepackage{hibib}
\hiBibConvert{<output-file>}
\addbibresource{<bib-file>}...
\begin{document}
\sentence{reference-1}.
\sentence{reference-2}.
\sentence{reference-3}.
...
\end{document}
```

It is necessary that the document cite every reference to be extracted from the `.bib` file. This is because `hibib` depends on the output of *biber*, which processes only references that were used in the document.

Once this document is constructed, compile the document, then run *biber* on it, and then compile it again. The `<output-file>` should now contain *Hereinafter*-style reference definitions corresponding to the references in the `.bib` file.

Note that this conversion process is incomplete and likely inaccurate, so you will probably have to fix many of the reference definitions manually.

```
\def\hiBibConvert#1{%
  \newwrite\tf@bibconv
  \immediate\openout\tf@bibconv #1.tex\relax
  \def\hi@bib@writeconv##1##2{%
    \begingroup
      ##1%
      \immediate\write\tf@bibconv{##2}%
    \endgroup
  }%
  \let\hi@bib@writeconv\@gobbletwo
```

6 Citation Command Compatibility

Biblatex uses different commands for inserting citations. Fundamentally there cannot be straightforward compatibility between those commands and *Hereinafter*, because the underlying data model of citations is different. Nevertheless, some partial compatibility is provided for the `\autocite` and `\autocites` commands, which could help with automatic conversion of Pandoc markdown documents.

```
\DeclareRobustCommand\autocites{%
  \hi@pse@acc@reset
  \hi@pse@bib@read
}
\let\autocite\autocites
\def\hi@pse@bib@read{%
  \futurelet\@let@token\hi@pse@bib@read@
}
\def\hi@pse@bib@read@{%
  \let\hi@pse@bib@suffix\@empty
```



```

\@testcase
\ifx\@let@token[\fi{\hi@pse@bib@read@oneopt}%
\ifx\@let@token\bgroup\fi{\hi@pse@bib@read@ref}%
\default{\hi@pse@bib@run}%
}
\def\hi@pse@bib@read@oneopt[#1]{%
\def\hi@pse@bib@suffix[#1]%
\futurelet\@let@token\hi@pse@bib@read@nextopt
}
\def\hi@pse@bib@read@nextopt{%
\test\ifx\@let@token[\fi{\hi@pse@bib@read@twoopts}{\hi@pse@bib@read@ref}%
}
\def\hi@pse@bib@read@twoopts[#1]{%
% Suffix is actually prefix
\expandafter\hi@pse@bib@parseprefix\hi@pse@bib@suffix~\@stop
\def\hi@pse@bib@suffix[#1]%
\hi@pse@bib@read@ref
}
\def\hi@pse@bib@parseprefix#1~#2\@stop{%
\hi@pse@acc@add\@gobble{#1}{}%
}
\def\hi@pse@bib@read@ref#1{%
\hi@pse@acc@addnonblank\@gobble{ }{}%
\hi@pse@acc@add\@gobble{#1}{}%
\hi@pse@acc@savework\hi@pse@svr
\ifx\hi@pse@bib@suffix\@empty
\hi@pse@acc@savcite
\else
\expandafter\hi@pse@state@page\expandafter{\hi@pse@bib@suffix}%
\fi
\hi@pse@bib@read
}
\def\hi@pse@bib@run{%
\unskip\footnote{%
\@expand{%
\hi@draw@citation{\@hi@captrue\@hi@sentrue}{\if@hi@dot\else.\fi}%
}\hi@pse@acc@all i%
}%
}

```